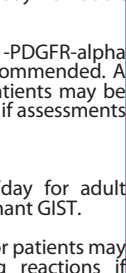


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IMINOX

Imatinib

NAME OF THE MEDICINE

IMINOX TABLET 100 MG
IMINOX TABLET 400 MG

COMPOSITION

Active Ingredient:

Iminox Tablet 100 mg

Each film-coated tablet contains imatinib mesilate equivalent to 100 mg imatinib.

Iminox Tablet 400 mg

Each film-coated tablet contains imatinib mesilate equivalent to 400 mg imatinib.

Excipients:

100 mg and 400 mg film-coated tablets

Povidone K-30 (Plasdone K-29/32), Magnesium Stearate, Opadry Brown 02F86982.

DESCRIPTION

Iminox Tablet 100 mg

Dark yellow to brownish orange colored, film coated tablets, round, biconvex with beveled edges debossed with 'S' and '1' on either side of break line on one side and plain on other side.

Iminox Tablet 400 mg

Dark yellow to brownish orange colored, film coated tablets, capsule shaped, biconvex with beveled edges debossed with 'S' and '2' on either side of break line on one side and plain on other side.

PHARMACODYNAMICS

Pharmacotherapeutic group: protein-tyrosine kinase inhibitor, ATC code: L01XE01.

Mechanism of action

Imatinib is a small molecule protein-tyrosine kinase inhibitor that potently inhibits the activity of the BCR-ABL tyrosine kinase (TK), as well as several receptor TKs: KIT, the receptor for stem cell factor (SCF) coded for by the c-KIT proto-oncogene, the discoidin domain receptors (DDR1 and DDR2), the colony stimulating factor receptor (CSF-1R) and the platelet-derived growth factor receptor alpha and beta (PDGFR-alpha and PDGFR beta). Imatinib can also inhibit cellular events mediated by activation of these receptor kinases.

Pharmacodynamic effects

Imatinib is a protein-tyrosine kinase inhibitor, which potently inhibits the breakpoint cluster region-Abelson (BCR-ABL) tyrosine kinase at the *in vitro*, cellular and *in vivo* levels. The compound selectively inhibits proliferation and induces apoptosis in BCR-ABL positive cell lines as well as fresh leukemic cells from Philadelphia chromosome positive CML and acute lymphoblastic leukemia (ALL) patients. In colony transformation assays using ex vivo peripheral blood and bone marrow samples, imatinib shows selective inhibition of BCR-ABL positive clones from CML patients.

In vivo the compound shows anti-tumour activity as a single agent in animal models using BCR-ABL positive tumour cells.

Imatinib is also an inhibitor of the receptor tyrosine kinases for platelet-derived growth factor (PDGFR) and stem cell factor (SCF), KIT, and inhibits PDGF- and SCF-mediated cellular events. *In vitro*, imatinib inhibits proliferation and induces apoptosis in gastrointestinal stromal tumour (GIST) cells, which express an activating KIT mutation. Constitutive activation of the PDGFR or the ABL protein tyrosine kinases as a consequence of fusion to diverse partner proteins or constitutive production of PDGF have been implicated in the pathogenesis of MDS/MPD, HES/CML and DFSP. In addition, constitutive activation of KIT or the PDGFR has been implicated in the pathogenesis of AML. Imatinib inhibits signaling and proliferation of cells driven by dysregulated PDGFR, KIT and ABL kinase activity.

Constitutive activation of KIT or the PDGFR has been implicated in the pathogenesis of ASM.

PHARMACOKINETICS

The pharmacokinetics of Iminox have been evaluated over a dosage range of 25 to 1,000 mg. Plasma pharmacokinetic profiles were analysed on day 1 and on either day 7 or day 28, by which time plasma concentrations had reached steady state.

Absorption

Mean absolute bioavailability for imatinib is 98%. The coefficient of variation for plasma imatinib AUC is in the range of 40% to 60% after an oral dose. When given with a high fat meal, the rate of absorption of imatinib was minimally reduced (11% decrease in C_{max} and prolongation of t_{max} by 1.5 h), with a small reduction in AUC (74%) compared to fasting conditions. The effect of gastrointestinal surgery on drug absorption has not been investigated.

Distribution

At relevant concentrations of imatinib, binding to plasma proteins was approximately 95% on the basis of *in vitro* experiments, mostly to albumin and alpha-acid-glycoprotein, with little binding to lipoprotein.

Metabolism

The main circulating metabolite in humans is the N-demethylated piperazine derivative, which shows similar *in vitro* potency to the parent. The plasma AUC for this metabolite was found to be only 16% of the AUC for imatinib. The plasma protein binding of the N-demethylated metabolite is similar to that of the parent compound.

Imatinib and the N-demethyl metabolite together accounted for about 65% of the circulating radioactivity ($AUC_{0-\infty}$). The remaining circulating radioactivity consisted of a number of minor metabolites.

The *in vitro* results showed that CYP3A4 was the major human P450 enzyme catalysing the biotransformation of imatinib. Of a panel of potential comedication (acetaminophen, acyclovir, allopurinol, amphotericin, cytarabine, erythromycin, fluconazole, hydroxyurea, norfloxacin, penicillin V) only erythromycin (IC₅₀ 50 µM) and fluconazole (IC₅₀ 118 µM) showed inhibition of imatinib metabolism which could have clinical relevance.

Imatinib was shown *in vitro* to be a competitive inhibitor of a marker substrates for CYP2C9, CYP2D6 and CYP3A4/5. Ki values in human liver microsomes were 27, 7.5 and 7.9 µmol/L, respectively. Maximal plasma concentrations of imatinib in patients are 2-4 µmol/L, consequently an inhibition of CYP2D6 and/or CYP3A4/5-mediated metabolism of co-administered drugs is possible. Imatinib did not interfere with the biotransformation of 5-fluorouracil, but it inhibited paclitaxel metabolism as a result of competitive inhibition of CYP2C8 (K_i = 34.7 µM). This Ki value is far higher than the expected plasma levels of imatinib in patients, consequently no interaction is expected upon co-administration of either 5-fluorouracil or paclitaxel and imatinib.

Elimination

Based on the recovery of compound(s) after an oral ¹⁴C-labelled dose of imatinib, approximately 81% of the dose was recovered within 7 days in faeces and urine. Unchanged imatinib accounted for 25% of the dose, the remainder being metabolites.

Plasma pharmacokinetics

The $t_{1/2}$ was approximately 18 h, suggesting that once-daily dosing is appropriate. The increase in mean AUC with increasing dose was linear and dose proportional in the range of 25 to 1,000 mg imatinib after oral administration. There was no change in the kinetics of imatinib on repeated dosing, and accumulation was 1.5 to 2.5-fold at steady state when dosed once daily.

Pharmacokinetics in GIST patients

In patients with GIST steady-state exposure was 1.5-fold higher than that observed for CML patients for the same dosage (400 mg daily). Based on population pharmacokinetic analysis in GIST patients, there were three variables (albumin, WBC and bilirubin) found to have a significant relationship with imatinib pharmacokinetics. Decreased values of albumin caused a reduced clearance (CL/f); and higher levels of WBC led to a reduction of CL/f, however, these associations are not sufficiently pronounced to warrant dose adjustment. In this patient population, the presence of hepatic metastases could potentially lead to hepatic insufficiency and reduced metabolism.

Population pharmacokinetics

Based on population pharmacokinetics analysis in CML patients, there was a small effect of age on the volume of distribution. This change is not thought to be clinically significant. The effect of body weight on the clearance of imatinib is such that for a patient weighing 50 kg the mean clearance is expected to be 8.5 L/h, while for a patient weighing 100 kg the clearance will rise to 11.8 L/h. These changes are considered sufficient to warrant dose adjustment based on kg bodyweight. There is no effect of gender on the kinetics of imatinib.

Pharmacokinetics in children

As in adult patients, imatinib was rapidly absorbed after oral administration in paediatric patients. Dosing in children at 260 and 340 mg/m²/day achieved the same exposure, respectively, as doses of 400 mg and 600 mg in adult patients. The comparison of AUC₀₋₂₄ on day 8 and day 1 at the 340 mg/m²/day dose level revealed a 1.7-fold drug accumulation after repeated once-daily dosing.

Based on population pharmacokinetic analysis in paediatric patients with haematological disorders (CML, Ph+ALL, or other haematological disorders treated with imatinib), clearance of imatinib increases with increasing body surface area (BSA). After correcting for the BSA effect, other demographics such as age, body weight and body mass index did not have clinically significant effects on the exposure of imatinib. The analysis confirmed that exposure of imatinib in paediatric patients receiving 260 mg/m² once daily (not exceeding 400 mg once daily) or 340 mg/m² once daily (not exceeding 600 mg once daily) were similar to those in adult patients who received imatinib 400 mg or 600 mg once daily.

Organ function impairment
Imatinib and its metabolites are not excreted via the kidney to a significant extent. Patients with mild and moderate impairment of renal function appear to have a higher plasma exposure than patients with normal renal function. The increase is approximately 1.5 to 2-fold, corresponding to a 1.5-fold elevation of plasma AGP, to which imatinib binds strongly. The free drug clearance of imatinib is probably similar between patients with renal impairment and those with normal renal function, since renal excretion represents only a minor elimination pathway for imatinib.

Although pharmacokinetic analysis showed that there is considerable inter subject variation, the mean exposure to imatinib did not increase in patients with varying degrees of liver dysfunction as compared to patients with normal liver function.

INDICATION

Imatinib is indicated for the treatment of :

- Adult and pediatric patients with newly diagnosed chronic myeloid leukemia (CML) as well as for the treatment of adult and pediatric patients with CML in blast crisis, accelerated phase, or in chronic phase after failure of interferon-alpha therapy.
- Adult patients with unresectable and/or metastatic malignant gastrointestinal stromal tumours (GIST).
- Adjuvant treatment of adult patients following resection of GIST. Patients who have a low or very low risk of recurrence should not receive adjuvant treatment.
- Adult and pediatric patients with newly diagnosed Philadelphia chromosome positive acute lymphoblastic leukemia (Ph+ ALL) integrated with chemotherapy.
- Adult patients with relapsed or refractory Ph+ ALL as monotherapy.
- Adult patients with myelodysplastic/myeloproliferative diseases (MDS/MPD) associated with platelet-derived growth factor receptor (PDGFR) gene rearrangements.
- Adult patients with hypereosinophilic syndrome (HES) and/or chronic eosinophilic leukemia (CEL) with FIP1L1-PDGFRa rearrangement.
- Adult patients with unresectable, recurrent and/or metastatic dermatofibrosarcoma protuberans (DFSP).
- Adult patients with aggressive systemic mastocytosis (ASM) without the D816V c-KIT mutation or with c-KIT mutational status unknown.

The effectiveness of imatinib is based on overall haematological and cytogenetic response rates and progression-free survival in CML, on haematological and cytogenetic response rates in Ph+ALL, MDS/MPD, on haematological response rates in HES/CML and ASM and on objective response rates in GIST and DFSP, and on recurrence-free survival in adjuvant GIST. The experience with imatinib in patients with MDS/MPD associated with PDGFR gene re-arrangements is very limited. Except in newly diagnosed chronic phase CML, there are no controlled trials demonstrating a clinical benefit or increased survival in diseases.

RECOMMENDED DOSAGE

DOSE REGIMEN AND ADMINISTRATION

Therapy should be initiated by a physician experienced in the treatment of patients with haematological malignancies and malignant sarcomas, as appropriate.

The prescribed dose should be administered orally with a meal and a large glass of water to minimize the risk of gastrointestinal disturbances. Doses of 400 mg or 600 mg should be administered once daily, whereas a daily dose of 800 mg should be administered as 400 mg twice a day, in the morning and in the evening.

For patients unable to swallow the film-coated tablets, the tablets may be dispersed in a glass of water or apple juice. The required number of tablets should be placed in the appropriate volume of beverage (approximately 50 mL for a 100 mg tablet, and 200 mL for a 400 mg tablet) and stirred with a spoon. The suspension should be administered immediately after complete disintegration of the tablets(s).

Treatment should be continued as long as the patient continues to benefit.

Monitoring of response to Imatinib therapy in Ph+ CML patients should be performed routinely and when therapy is modified, to identify suboptimal response, loss of response to therapy, poor patient compliance, or possible drug-drug interaction. Results of monitoring should guide appropriate CML management.

General target population:

Dosage in CML

The recommended dose of Imatinib is 400 mg/day for adult patients in chronic phase CML and 600 mg/day for patients in accelerated phase or blast crisis.

Dose increase from 400 mg to 600 mg or 800 mg in patients with chronic phase disease, or from 600 mg to a maximum of 800 mg daily in patients in accelerated phase or blast crisis may be considered in the absence of severe adverse drug reaction and severe non-leukaemia-related neutropenia or thrombocytopenia in the following circumstances: disease progression (at any time); failure to achieve a satisfactory haematological response after at least 3 months of treatment; failure to achieve a cytogenetic response after 12 months of treatment; or loss of a previously achieved haematological and/or cytogenetic response.

See section on special populations for paediatric patients.

Dosage in Ph+ ALL

The recommended dose of Imatinib is 600 mg/day for adult patients with Ph+ ALL.

See section on special populations for paediatric patients.

Dosage in MDS/MPD

The recommended dose of Imatinib is 400 mg/day for adult patients with MDS/MPD.

Dosage in ASM

The recommended dose of Imatinib is 400 mg/day for adult patients with ASM without the D816V c-KIT mutation or mutational status unknown or not responding satisfactorily to other therapies.

For patients with ASM associated with eosinophilia, a clonal haematological disease related to the fusion kinase FIP1L1-PDGFR-alpha, a starting dose of 100 mg/day is recommended. A dose increase from 100 mg to 400 mg for these patients may be considered in the absence of adverse drug reactions if assessments demonstrate an insufficient response to therapy.

Dosage in HES/CEL

The recommended dose of Imatinib is 400 mg/day for adult patients with HES/CEL.

For HES/CEL patients with demonstrated FIP1L1-PDGFR-alpha fusion kinase, a starting dose of 100 mg/day is recommended. A dose increase from 100 mg to 400 mg for these patients may be considered in the absence of adverse drug reactions if assessments demonstrate an insufficient response to therapy.

Dosage in GIST

The recommended dose of Imatinib is 400 mg/day for adult patients with unresectable and/or metastatic, malignant GIST.

A dose increase from 400 mg to 600 mg or 800 mg for patients may be considered in the absence of adverse drug reactions if assessments demonstrate an insufficient response to therapy.

The recommended dose of Imatinib is 400 mg/day for the adjuvant treatment of adult patients following complete gross resection of GIST. The optimal treatment duration with Imatinib is not known.

Dosage in DFSP

The recommended dose of Imatinib is 800 mg/day for adult patients with DFSP.

Dose adjustments for adverse drug reactions

Non-haematological adverse drug reactions

If a severe non-haematological adverse drug reaction develops with Imatinib use, treatment must be withheld until the event has resolved. Thereafter, treatment can be resumed as appropriate depending on the initial severity of the event.

If elevations in bilirubin >3 x institutional upper limit of normal (IULN) or in liver transaminases >5 x IULN occur, Imatinib should be withheld until bilirubin levels have returned to a <1.5 x IULN and transaminase levels to <2.5 x IULN. Treatment with Imatinib may then be continued at a reduced daily dose. In adults the dose should be reduced from 400 to 300 mg, or from 600 to 400 mg, or from 800 mg to 600 mg, and in paediatric patients from 340 to 260 mg/m²/day.

Hematological adverse drug reactions

Dose reduction or treatment interruption for severe neutropenia and thrombocytopenia are recommended as indicated in the table below.

Table 1: Dose adjustments for neutropenia and thrombocytopenia:		
ASM associated with eosinophilia and HES/CEL with FIP1L1-PDGFR-alpha fusion kinase (starting dose 100 mg)	ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L	1. Stop imatinib until ANC ≥ 1.5 x10 ⁹ /L and platelets ≥ 75 x10 ⁹ /L. 2. Resume treatment with imatinib at previous dose (i.e. before severe adverse drug reaction).
Chronic phase CML, MDS/MPD, ASM, HES/CEL and/or ALL (starting dose 400 mg)	ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L	1. Stop imatinib until ANC ≥ 1.5 x10 ⁹ /L and platelets ≥ 75 x10 ⁹ /L. 2. Resume treatment with imatinib at previous dose (i.e. before severe adverse drug reaction). 3. In the event of recurrence of ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L, repeat step 1 and resume imatinib at reduced dose of 300 mg.
Pediatric chronic phase CML (at dose 340 mg/m ²)	ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L	1. Stop imatinib until ANC ≥ 1.5 x10 ⁹ /L and platelets ≥ 75 x10 ⁹ /L. 2. Resume treatment with imatinib at previous dose (i.e. before severe adverse drug reaction). 3. In the event of recurrence of ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L, repeat step 1 and resume imatinib at reduced dose of 260 mg/m ² .
Accelerated phase CML and blast crisis and Ph+ ALL (starting dose 600 mg)	ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L	1. Check whether cytopenia is related to leukemia (marrow aspirate or biopsy). 2. If cytopenia is unrelated to leukemia, reduce dose of imatinib to 400 mg ^a . 3. If cytopenia persists for 2 weeks, reduce further to 300 mg ^a . 4. If cytopenia persists for 4 weeks and is still unrelated to leukemia, stop imatinib until ANC ≥ 1 x10 ⁹ /L and platelets ≥ 20 x10 ⁹ /L, then resume treatment at 300 mg ^a .
DFSP (starting dose 800 mg)	ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L	1. Stop imatinib until ANC ≥ 1.5 x10 ⁹ /L and platelets ≥ 75 x10 ⁹ /L. 2. Resume treatment with imatinib at 600 mg. 3. In the event of recurrence of ANC < 1.0 x10 ⁹ /L and/or platelets < 50 x10 ⁹ /L, repeat step 1 and resume imatinib at reduced dose of 400 mg.

ANC = absolute neutrophil count

^a occurring after at least 1 month of treatment

^b or 260 mg/m² in children

^c or 340 mg/m² in children

^d or 200 mg/m² in children

Special populations

Renal insufficiency

Imatinib and its metabolites are not significantly excreted via the kidney. Patients with renal dysfunction or on dialysis could be given the minimum recommended dose of 400 mg daily as starting dose. However, in these patients caution is recommended. The dose can be reduced if not tolerated. If tolerated, the dose can be increased for lack of efficacy.

Hepatic impairment

Imatinib is mainly metabolized by the liver. Patients with mild, moderate or severe liver impairment should be given the minimum recommended dose of 400 mg daily. The dose can be reduced if not tolerated.

Paediatric patients (below 18 years)

There is no experience with the use of Imatinib in paediatric patients with CML below 2 years of age and with Ph+ALL below 1 year of age. There is very limited to no experience with the use of imatinib in paediatric patients in other indications.

Dosing in paediatric patients should be on the basis of body surface area (mg/m²). The dose of 340 mg/m² daily is recommended for children with chronic phase and advanced phase CML and Ph+ALL (not to exceed the total dose of 600 mg daily). Treatment can be given as a once daily dose in CML and Ph+ALL. In CML, alternatively the daily dose may be split into two administrations – one in the morning and one in the evening.

Geriatric patients (65 years or above)

No significant age-related pharmacokinetic differences have been observed in adult patients which included patients age 65 and older. No specific dose recommendation is necessary in the elderly.

ROUTE OF ADMINISTRATION

Oral

CONTRAINDICATIONS

Hypersensitivity to the active substance or to any of the excipients used in the composition.

WARNING AND PRECAUTIONS

When imatinib is co-administered with other medicinal products, there is a potential for drug interactions. Caution should be used when taking imatinib with protease inhibitors, azole antifungals, certain macrolides, CYP3A4 substrates with a narrow therapeutic window (e.g. cyclosporine, pimozide, tacrolimus, sirolimus, ergotamine, diergotamine, fentanyl, alfentanil, terfenadine, bortezomib, docetaxel, quinidine) or warfarin and other coumarin derivatives.

Concomitant use of imatinib and medicinal products that induce CYP3A4 (e.g. dexamethasone, phenytoin, carbamazepine, rifampicin, phenobarbital or *Hypericum perforatum*, also known as St. John's Wort) may significantly reduce exposure to imatinib, potentially increasing the risk of therapeutic failure. Therefore, concomitant use of strong CYP3A4 inducers and imatinib should be avoided.

Hypothyroidism

Thyroid-Stimulating Hormone (TSH) levels should be closely monitored in thyroidectomy patients undergoing levothyroxine replacement during treatment with imatinib.

Hepatotoxicity

Metabolism of imatinib is mainly hepatic, and only 13% of excretion is through the kidneys. In patients with hepatic dysfunction (mild, moderate or severe), peripheral blood counts and liver enzymes should be carefully monitored. It should be noted that GIST patients may have hepatic metastases which could lead to hepatic impairment.

Cases of liver injury, including hepatic failure, hepatic necrosis, have been observed with imatinib. When imatinib is combined with high dose chemotherapy regimens, an increase in serious hepatic reactions has been detected. Hepatic function should be carefully monitored in circumstances where imatinib is combined with chemotherapy regimens also known to be associated with hepatic dysfunction.

Fluid Retention

Occurrences of severe fluid retention (pleural effusion, oedema, pulmonary oedema, ascites, and superficial oedema) have been reported in newly diagnosed CML patients taking imatinib. Therefore, it is highly recommended that patients be weighed regularly. An unexpected rapid weight gain should be carefully investigated and if necessary appropriate supportive care and therapeutic measures should be undertaken. There was an increased incidence of these events in older people and those with a prior history of cardiac disease. Therefore, caution should be exercised in patients with cardiac dysfunction.

Patients with Cardiac Disease or Renal Failure

Patients with cardiac disease, risk factors for cardiac failure or history of renal failure should be monitored carefully, and any patients with signs or symptoms consistent with cardiac or renal failure should be evaluated and treated.

In patients with hypereosinophilic syndrome (HES) with occult infiltration of HES cells within the myocardium, isolated cases of cardiogenic shock/left ventricular dysfunction have been associated with HES cell degeneration upon the initiation of imatinib therapy. The condition was reported to be reversible with the administration of systemic steroids, circulatory support measures and temporarily withholding imatinib. As cardiac events have been reported uncommonly with imatinib, a careful assessment of the benefit/risk of imatinib therapy should be considered in HES/CML population before treatment initiation. Myelodysplastic (MDS)/myeloproliferative diseases (MPD) with PDGFR gene re-arrangements could be associated with high eosinophil levels. Evaluation by cardiology specialist, performance of an echocardiogram and determination of serum troponin should therefore be considered in patients with HES/CEL, and in patients with MDS/MPD associated with high eosinophil levels before imatinib is administered. If either is abnormal, follow up with a cardiology specialist and the prophylactic use of systemic steroids (1 to 2 mg/kg) for one to two weeks concomitantly with imatinib should be considered at the initiation of therapy.

Gastrointestinal Haemorrhage

Both gastrointestinal and intra-tumoural haemorrhages were reported in patients with unresectable and/or metastatic GIST. No predisposing factors (e.g. tumour size, tumour location, coagulation disorders) have been identified that place patients with GIST at a higher risk of either type of haemorrhage. Since increased vascularity and propensity for bleeding is a part of the nature and clinical course of GIST, standard practices and procedures for the monitoring and management of haemorrhage in all patients should be applied.

In addition, gastric antral vascular ectasia (GAVE), a rare cause of gastrointestinal haemorrhage, has been reported in post-market experience in patients with CML, ALL and other diseases. When needed, discontinuation of imatinib treatment may be considered.

Tumor Lysis Syndrome

Due to the possible occurrence of tumour lysis syndrome (TLS), correction of clinically significant dehydration and treatment of high uric acid levels are recommended prior to initiation of imatinib.

Hepatitis B Reactivation

Reactivation of hepatitis B in patients who are chronic carriers of this virus has occurred after these patients received BCR-ABL tyrosine kinase inhibitors. Some cases resulted in acute hepatic failure or fulminant hepatitis leading to liver transplantation or a fatal outcome.

Patients should be tested for HBV infection before initiating treatment with imatinib. Experts in liver disease and in the treatment of hepatitis B should be consulted before treatment is initiated in patients with positive hepatitis B serology (including those with active disease) and for patients who test positive for HBV infection during treatment. Carriers of HBV who require treatment with imatinib should be closely monitored for signs and symptoms of active HBV infection throughout therapy and for several months following termination of therapy.

Phototoxicity

Exposure to direct sunlight should be avoided or minimised due to the risk of phototoxicity associated with imatinib treatment. Patients should be instructed to use measures such as protective clothing and sunscreen with high sun protection factor (SPF).

Thrombotic Microangiopathy

BCR-ABL tyrosine kinase inhibitors (TKIs) have been associated with thrombotic microangiopathy (TMA), including individual case reports for imatinib. If laboratory or clinical findings associated with TMA occur in a patient receiving imatinib, treatment should be discontinued and thorough evaluation for TMA, including ADAMTS13 activity and anti-ADAMTS13-antibody determination, should be completed. If anti-ADAMTS13-antibody is elevated in conjunction with low ADAMTS13 activity, treatment with imatinib should not be resumed.

Laboratory Tests

Complete blood counts must be performed regularly during therapy with imatinib. Treatment of CML patients with imatinib has been associated with neutropenia or thrombocytopenia. However, the occurrence of the cytopenias is likely to be related to the stage of disease being treated and they were more frequent in patients with accelerated phase CML or blast crisis as compared to patients with chronic phase CML. Treatment with imatinib may be interrupted or the dose may be reduced, as recommended in section Recommended Dose.

Liver function (transaminases, bilirubin, alkaline phosphatase) should be monitored regularly in patients receiving imatinib.

In patients with impaired renal function, imatinib plasma exposure seems to be higher than that in patients with normal renal function, probably due to an elevated plasma level of alpha-acid glycoprotein (AGP), an imatinib-binding protein, in these patients. Patients with renal impairment should be given the minimum starting dose. Patients with severe renal impairment should be treated with caution. The dose can be reduced if not tolerated.

Long-term treatment with imatinib may be associated with a clinically significant decline in renal function. Renal function should, therefore, be evaluated prior to the start of imatinib therapy and closely monitored during therapy, with particular attention to those patients exhibiting risk factors for renal dysfunction. If renal dysfunction is observed, appropriate management and treatment should be prescribed in accordance with standard treatment guidelines.

Paediatric Population

There have been case reports of growth retardation occurring in children and pre-adolescents receiving imatinib. In the CML paediatric population, a significant decrease (but of uncertain clinical relevance) in median height standard deviation was reported irrespective of pubertal status or gender. Close monitoring of growth in children under imatinib treatment is recommended.

INTERACTIONS WITH OTHER MEDICAMENTS

Drugs that may increase imatinib plasma concentrations: Substances that inhibit the cytochrome P450 isoenzyme CYP3A4 activity (e.g. protease inhibitors such as indinavir, lopinavir/ritonavir, ritonavir, saquinavir, telaprevir, nelfinavir, boceprevir; azole antifungals including ketoconazole, itraconazole, posaconazole, voriconazole; certain macrolides such as erythromycin, clarithromycin and telithromycin) could decrease metabolism and increase imatinib concentrations. There was a significant increase in exposure to imatinib (the mean C_{max} and AUC of imatinib) when it was co-administered with a single dose of ketoconazole (a CYP3A4 inhibitor). Caution should be taken when administering imatinib with inhibitors of the CYP3A4 family.

Drugs that may decrease imatinib plasma concentrations: Substances that are inducers of CYP3A4 activity (e.g. dexamethasone, phenytoin, carbamazepine, rifampicin, phenobarbital, fosphenytoin, pirimidone or *Hypericum perforatum*, also known as St. John's Wort) may significantly reduce exposure to imatinib, potentially increasing the risk of therapeutic failure. Pretreatment with multiple doses of rifampicin 600 mg followed by a single 400 mg dose of imatinib resulted in decrease in C_{max} and AUC(0-∞), of the respective values without rifampicin treatment. The plasma AUC for imatinib decreased by 73% in patients with malignant gliomas treated with imatinib while taking enzyme-inducing anti-epileptic drugs (EIAEDs) such as carbamazepine, oxcarbazepine and phenytoin. Concomitant use of rifampicin or other strong CYP3A4 inducers and imatinib should be avoided.

Drugs that may have their plasma concentration altered by imatinib Imatinib increases the mean C_{max} and AUC of simvastatin (CYP3A4 substrate) 2- and 3.5-fold, respectively, indicating an inhibition of the CYP3A4 by imatinib. Therefore, caution is recommended when administering imatinib with CYP3A4 substrates with a narrow therapeutic window (e.g. cyclosporine, pimozide, tacrolimus, sirolimus, ergotamine, diergotamine, fentanyl, alfentanil, terfenadine, bortezomib, docetaxel and quinidine). Imatinib may increase plasma concentration of other CYP3A4 metabolised drugs (e.g. triazolo-benzodiazepines, dihydropyridine calcium channel blockers, certain HMG-CoA reductase inhibitors i.e. statins, etc.).

Because of known increased risks of bleeding in conjunction with the use of imatinib (e.g. haemorrhage), patients who require anticoagulation should receive low-molecular-weight or standard heparin, instead of coumarin derivatives such as warfarin.

In vitro, imatinib inhibits the cytochrome P450 isoenzyme CYP2D6 activity at concentrations similar to those that affect CYP3A4 activity. Imatinib at 400 mg twice daily had an inhibitory effect on CYP2D6-mediated metoprolol metabolism, with metoprolol C_{max} and AUC being increased by approximately 23%. Co-administration of imatinib with CYP2D6 substrates does not seem to be a risk factor for drug-drug interactions and dose adjustment may not be necessary. However, caution is advised for CYP2D6 substrates with a narrow therapeutic window such as metoprolol. In patients treated with metoprolol clinical monitoring should be considered.

In vitro, imatinib inhibits paracetamol O-glucuronidation (Ki value of 58.5 micromol/L). This inhibition has not been observed in vivo after the administration of imatinib 400 mg and paracetamol 1000 mg. Higher doses of imatinib and paracetamol have not been studied.

Caution should therefore be exercised when using high doses of imatinib and paracetamol concomitantly.

In thyroidectomy patients receiving levothyroxine, the plasma exposure to levothyroxine may be decreased when imatinib is co-administered. Caution is therefore recommended. However, the mechanism of the observed interaction is presently unknown.

In Ph+ ALL patients, there is clinical experience of co-administering imatinib with chemotherapy, but drug-drug interactions between imatinib and chemotherapy regimens are not well characterized. Imatinib adverse events, i.e. hepatotoxicity, myelosuppression or others, may increase and it has been reported that concomitant use with L-asparaginase could be associated with increased hepatotoxicity. Therefore, the use of imatinib in combination requires special precaution.

PREGNANCY AND LACTATION

Women of childbearing potential Women of childbearing potential must be advised to use effective contraception during treatment and for at least 15 days after stopping treatment with imatinib.

Pregnancy

There are limited data on the use of imatinib in pregnant women. There have been post-marketing reports of spontaneous abortions and infant congenital anomalies from women who have taken imatinib. Imatinib should not be used during pregnancy unless clearly necessary. If it is used during pregnancy, the patient must be informed of the potential risk to the fetus.

Breast-feeding

There is limited information on imatinib distribution on human milk. Since the effects of low-dose exposure of the infant to imatinib are unknown, women should not breast-feed during treatment and for at least 15 days after stopping treatment with imatinib.

Fertility

Patients concerned about their fertility on imatinib treatment should consult with their physician.

ADVERSE EFFECTS

Patients with advanced stages of malignancies may have numerous confounding medical conditions that make causality of adverse reactions difficult to assess due to variety of symptoms related to the underlying disease, its progression, and the co-administration of numerous medicinal products.

The adverse reactions were similar in all indications, with two exceptions. There was more myelosuppression seen in CML patients than in GIST, which is probably due to the underlying disease. The most commonly reported drug-related adverse reactions in both settings were mild nausea, vomiting, diarrhoea, abdominal pain, fatigue, myalgia, muscle cramps and rash. Superficial oedemas were a common finding and were described primarily as periorbital or lower limb oedemas. However, these oedemas were rarely severe and may be managed with diuretics, other supported measures, or by reducing the dose of imatinib.

When imatinib was combined with high dose chemotherapy in Ph+ ALL patients, transient liver toxicity in the form of transaminase elevation and hyperbilirubinaemia were observed. Considering the limited safety database, the adverse events thus far reported in children are consistent with the known safety profile in adult patients with Ph+ ALL. The safety database for children with Ph+ ALL is very limited though no new safety concerns have been identified.

Miscellaneous adverse reactions such as pleural effusion, ascites, pulmonary edema and rapid weight gain with or without superficial edema may be collectively described as "fluid retention". These reactions can usually be managed by withholding imatinib temporarily and with diuretics and other appropriate supportive care measures. However, some of these reactions may be serious or life-threatening and several patients with blast crisis died with a complex clinical history of pleural effusion, congestive heart failure and renal failure.

Adverse reactions

Adverse reactions reported as more than an isolated case as listed below, by system organ class and by frequency.

Infections and infestations	
Uncommon	Herpes zoster, herpes simplex, nasopharyngitis, pneumonia ¹ , sinusitis, cellulitis, upper respiratory tract infection, influenza, urinary tract infection, gastroenteritis, sepsis
Rare	Fungal infection
Not known	Hepatitis B reactivation ²
Neoplasm benign, malignant and unspecified (including cysts and polyps)	
Rare	Tumour lysis syndrome
Not known	Tumour haemorrhage/tumour necrosis ³
Immune system disorders	
Not known	Anaphylactic shock ⁴
Blood and lymphatic system disorders	
Very common	Neutropenia, thrombocytopenia, anaemia
Common	Pancytopenia, febrile neutropenia
Uncommon	Thrombocythaemia, lymphopenia, bone marrow depression, eosinophilia, lymphadenopathy
Rare	Haemolytic anaemia, thrombotic microangiopathy
Metabolism and nutrition disorders	
Common	Anorexia
Uncommon	Hypokalaemia, increased appetite, hypophosphataemia, decreased appetite, dehydration, gout, hyperuricaemia, hypercalcaemia, hyperglycaemia, hyponatraemia
Rare	Hyperkalaemia, hypomagnesaemia
Psychiatric disorders	
Common	Insomnia
Uncommon	Depression, libido decreased, anxiety
Rare	Confusional state
Nervous system disorders	
Very common	Headache ⁵
Common	Dizziness, paraesthesia, taste disturbance, hypoaesthesia
Uncommon	Migraine, somnolence, syncope, peripheral neuropathy, memory impairment, sciatia, restless leg syndrome, tremor, cerebral haemorrhage
Rare	Increased intracranial pressure, convulsions, optic neuritis
Not known	Cerebral oedema ⁶
Eye disorders	
Common conjunctival	Eye lid oedema, lacrimation increased, haemorrhage, conjunctivitis, dry eye, blurred vision
Uncommon	Eye irritation, eye pain, orbital oedema, scleral haemorrhage, retinal haemorrhage, blepharitis, macular oedema
Rare	Cataract, glaucoma, papilloedema
Not known	Vitreous haemorrhage ⁸
Ear and labyrinth disorders	
Uncommon	Vertigo, tinnitus, hearing loss
Cardiac disorders	
Common	Palpitations, tachycardia, cardiac failure, congestive ⁹
Rare	Arrhythmia, atrial fibrillation, cardiac arrest, myocardial infarction, angina pectoris, pericardial effusion
Not known	Pericarditis ⁹ , cardiac tamponade ⁸
Vascular disorders⁴	
Common	Flushing, haemorrhage
Uncommon	Hypertension, haematoma, subdural haematoma, peripheral coldness, hypotension, Raynaud's phenomenon
Not known	Thrombosis/embolism ⁸
Respiratory, thoracic and mediastinal disorders	
Common	Dyspnoea, epistaxis, cough
Uncommon	Pleural effusion ⁷ , pharyngolaryngeal pain, pharyngitis
Rare	Pleuritic pain, pulmonary fibrosis, pulmonary pulmonary haemorrhage ¹¹
hypertension, Not known	Acute respiratory failure ¹¹ , interstitial lung disease ⁸
Gastrointestinal disorders	
Very common	Nausea, diarrhoea, vomiting, dyspepsia, abdominal pain ⁶
Common	Flatulence, abdominal distension, gastro-oesophageal reflux, constipation, dry mouth, gastritis
Uncommon	Stomatitis, mouth ulceration, gastrointestinal haemorrhage ⁷ , eructation, melaena, oesophagitis, ascites, gastric ulcer, haematemesis, cheilitis, dysphagia, pancreatitis
Rare	Colitis, ileus, inflammatory bowel disease
Not known	Ileus/intestinal obstruction ⁷ , gastrointestinal perforation ⁷ , diverticulitis ⁷ , gastric antral vascular ectasia (GAVE) ⁷
Hepatobiliary disorders	
Common	Increased hepatic enzymes
Uncommon	Hyperbilirubinaemia, hepatitis, jaundice
Rare	Hepatic failure ⁸ , hepatic necrosis

Skin and subcutaneous tissue disorders

Very common	Periorbital oedema, dermatitis/eczema/rash
Common	Pruritus, face oedema, dry skin, erythema, alopecia, night sweats, photosensitivity reaction
Uncommon	Rash pustular, contusion, sweating increased, urticaria, ecchymosis, increased tendency to bruise, hypotrichosis, skin hypopigmentation, dermatitis exfoliative, onychoclasia, folliculitis, petechiae, psoriasis, purpura, skin hyperpigmentation, bullous eruptions
Acute febrile neutrophilic dermatosis (Sweet's syndrome), nail discolouration, angioneurotic oedema, rash vesicular, erythema multiforme, leucocytoclastic vasculitis, Stevens-Johnson syndrome, acute generalized exanthematous pustulosis (AGEP)	
Rare	Palmoplantar erythrodysesthesia syndrome ⁸ , lichenoid keratosis ⁸ , lichen planus ⁸ , toxic epidermal necrolysis ⁸ , drug rash with eosinophilia and systemic symptoms (DRESS) ⁸ , pseudoporphyria ⁸
Not known	
Musculoskeletal and connective tissue disorders	
Very common including	Muscle spasm and cramps, musculoskeletal pain
Common	myalgia ⁸ , arthralgia, bone pain ¹⁰
Uncommon	Joint swelling
Uncommon	Joint and muscle stiffness
Rare	Muscular weakness, arthritis, rhabdomyolysis/myopathy
Not known	Avascular necrosis/hip necrosis ⁸ , growth retardation in children ⁸
Renal and urinary disorders	
Uncommon	Renal pain, haematuria, renal failure acute, urinary frequency decreased
Not known	Renal failure chronic
Reproductive system and breast disorders	
Uncommon	Gynaecomastia, erectile dysfunction, menorrhagia, menstruation irregular, sexual dysfunction, nipple pain, breast enlargement, scrotal oedema
Rare	Haemorrhagic corpus luteum/haemorrhagic ovarian cyst
General disorders and administration site conditions	
Very common	Fluid retention and oedema, fatigue
Common	Weakness, pyrexia, anasarca, chills, rigors
Uncommon	Chest pain, malaise
Investigations	
Very common	Weight increased
Common	Weight decreased
Uncommon	Blood creatinine increased, blood creatine phosphokinase increased, blood lactate dehydrogenase increased, blood alkaline phosphatase increased
Rare	Blood amylase increased

⁸ These types of reactions have been reported mainly from post-marketing experience with imatinib. This includes spontaneous case reports as well. Because these reactions are reported from a population of uncertain size, it is not always possible to reliably estimate their frequency or establish a causal relationship to imatinib exposure.

¹⁰ Pneumonia was reported most commonly in patients with transformed CML and in patients with GIST.

² Headache was the most common in GIST patients.

³ On a patient-year basis, cardiac events including congestive heart failure were more commonly observed in patients with transformed CML than in patients with chronic CML.

⁴ Flushing was most common in GIST patients and bleeding (hematoma, haemorrhage) was most common in patients with GIST and with transformed CML (CML-AP and CML-BC).

⁵ Pleural effusion was reported more commonly in patients with GIST and in patients with transformed CML (CML-AP and CML-BC) than in patients with chronic CML.

^{6,7} Abdominal pain and gastrointestinal haemorrhage were most commonly observed in GIST patients.

⁸ Some fatal cases of hepatic failure and of hepatic necrosis have been reported.

⁹ Musculoskeletal pain during treatment with imatinib or after discontinuation has been observed in post-marketing.

¹⁰ Musculoskeletal pain and related events were more commonly observed in patients with CML than in GIST patients.

¹¹ Fatal cases have been reported in patients with advanced disease, severe infections, severe neutropenia and other serious concomitant conditions.

Description of selected adverse reactions

Hepatitis B reactivation

Hepatitis B reactivation has been reported in association with BCR-ABL TKIs. Some cases resulted in acute hepatic failure of fulminant hepatitis leading to liver transplantation or a fatal outcome.

OVERDOSAGE

Experience with doses higher than the recommended therapeutic dose is limited. Isolated cases of imatinib overdose have been reported spontaneously and in the literature. In the event of overdose the patient should be observed and appropriate symptomatic treatment given. Generally, the reported outcome in these cases was "improved" or "recovered". Events that have been reported at different dose ranges as follows:

Adult population:

1,200 to 1,600 mg (duration varying between 1 to 10 days): Nausea, vomiting, diarrhoea, rash, erythema, oedema, swelling, fatigue, muscle spasms, thrombocytopenia, pancytopenia, abdominal pain, headache, decreased appetite.

1,800 to 3,200 mg (as high as 3,200 mg daily for 6 days): Weakness, myalgia, increased creatinine phosphokinase, increased bilirubin, gastrointestinal pain.

6,400 mg (single dose): One case in the literature reported one patient who experienced nausea, vomiting, abdominal pain, pyrexia, facial swelling, neutrophil count decreased, increased transaminases.

8 to 10 g (single dose): Vomiting and gastrointestinal pain have been reported.

Pediatric population:

One 3-year-old male exposed to a single dose of 400 mg experienced vomiting, diarrhoea and anorexia and another 3-year-old male exposed to a single dose of 980 mg dose experienced decreased white blood cell count and diarrhoea.

In the event of overdose, the patient should be observed and appropriate supportive treatment given.

DRIVING AND USE MACHINES

Patients should be advised that they may experience undesirable effects such as dizziness, blurred vision or somnolence during treatment with imatinib. Therefore, caution should be recommended when driving a car or operating machinery.

STORAGE CONDITIONS

Store below 30°C.

Protect from moisture.

Store in original package.

SHELF LIFE

3 years

PACKAGING

IMINOX TABLET 100 MG : Pack of 60 tablets (6 blisters x 10 tablets)

IMINOX TABLET 400 MG : Pack of 30 tablets (3 blisters x 10 tablets)

MANUFACTURER

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